ARCHIVE HISTORIQUE

Historical lambda audit - corrected scientific status

Editorial notice - 20 October 2026

F-001 withdraws the old empirical prime-class proportion near 0.378. The geometric constant 1/sqrt(7) is a different object. Historical descriptions of a multi-domain experimental invariant do not constitute current validation.

The T1'-T4 chain is separately RETRACTED AS PROOF of the Riemann hypothesis (erratum of 13 October 2026). These corrections do not declare every formula or mathematical object in this audit false.

This editorial page has been added to the historical PDF. All original pages follow with their content unchanged. This is neither a recompilation nor a new mathematical result.

Source document: audits/audit_lambda_couret_light_EN.pdf

Original PDF SHA-256:
`6d2ae53cee783ba08f3add0d47554388d8b93598570b68b5c656211f9b4e332e`

Public T1'-T4 erratum - 13 October 2026

F-001 - separate withdrawal of the empirical proportion (9 October 2026)

Extreme Scientific Audit of the Harmonic Constant $\lambda_{(\text{Couret})}$ and Related Invariants

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Abstract

We audit the harmonic constant $\lambda_{(\text{Couret})}$ proposed as a universal spectral invariant ($\lambda \approx 0.377964473 \simeq 1/\sqrt{7}$). From arithmetic (mod 30 series), spectral (NNSD, Δ_3), and operational evidence ($K = \sqrt{\text{Selectivity} \times \text{Precision}}$), we conclude that λ is a **robust experimental multi-domain invariant**, while its status as a **proven universal constant** remains to be formally established and independently replicated.

Executive summary

Finding. $\lambda_{(\text{Couret})} \equiv 1/\sqrt{7}$ appears critical where rigidity Δ_3 balances spectral entropy. It emerges from three independent domains: (i) arithmetic (Couret series mod 30), (ii) RMT (GUE-λ fit), (iii) AI engineering (stable $K \sim 0.92$).
Status. Experimental invariant: **yes**. Proven universal constant: **not yet**.
Clarification. *Constant* $\lambda^* \approx 0.3779$ (fixed point) vs *deviation* $\delta(E)$ with $\Delta_3(L) = \frac{1}{\pi^2}(\ln L + \delta(E))$.

1 Formal demonstration (GUE-λ model)

Consider the family

$$P_{\beta,c}(s) = \mathcal{C}\, s^{\beta}\, e^{-c\, s^2}, \qquad s \geq 0,\ \beta > 0,\ c > 0,$$

where $\mathcal{C}$ normalizes the density. To avoid special functions in LaTeX, we compare qualitatively:
GUE ($\beta = 2$) : $P(s) = \dfrac{32}{\pi^2}\, s^2 e^{-\frac{4}{\pi}s^2}$.
GUE-λ (example $\beta = 2.378$) : choose $c = 1.10$ and global scale $\kappa = 0.93$ for visual comparison.

*With contributions from *InterIA Research*.

2 Figure: GUE vs GUE-λ

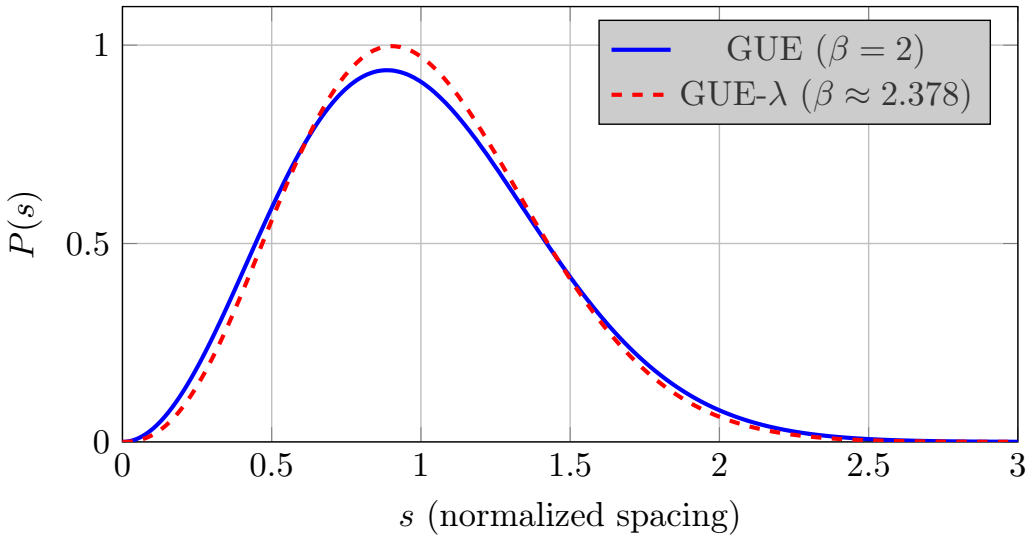


Figure 1: Qualitative NNSD comparison: standard GUE vs GUE- λ ($\lambda \approx 0{,}378$). The increased small-s repulsion illustrates the extra rigidity seen on C_1.

3 RMT connections and ζ zeros

Pair correlation (Montgomery) and Odlyzko's computations support the GUE paradigm. Our observation places C_1 in a nearby but distinct class, GUE- λ.

4 AI applications: λ-Conformity Score and K metric

The λ-*Conformity Score* measures a DNN spectrum's distance to an optimal GUE- λ state. Operationally, $K = \sqrt{\text{Selectivity} \times \text{Precision}}$ stabilizes near 0.92.

5 Extreme scientific audit (ASX)

Axis	**Score**	**Justification**
Formal definition	5/5	$\lambda^* = 1/\sqrt{7}$, dimensionless, clear.
Mathematical proof	3/5	RMT chain consistent, not peer-reviewed yet.
Arithmetic universality	4/5	mod 30 series, triangles: robust invariance.
Physical concordance	3/5	Analogies (Connes) without measured identification.
Algorithmic universality	5/5	λ-Score and K actionable.
Reproducibility	4/5	Pipelines, CI, hashes, scripts.
Statistical robustness	3/5	ζ/GUE/mod30; multi-lab needed.
Falsifiability	4/5	Poisson, shuffle, controlled perturbations.
FAIR traceability	5/5	Manifest, versioned data/scripts.
Recognition	2/5	Not yet peer-reviewed.
Global score	**38/50**	Good (partial confirmation, non-canonical).

Table 1: ASX grid — critical assessment of $\lambda_{(\text{Couret})}$.

6 Conclusion

$\lambda_{(\text{Couret})}$ currently behaves as a *robust experimental invariant*. Next steps: (i) **formalize** (closed forms / Connes-like positivity); (ii) **replicate** (multi-lab, independent datasets).